JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU& MSBTE Mumbai
Department of Computer Applications
"Powered by Progress , Inspired by Vision"
Session 2026-27

VISION	MISSION
"To be the source of bringing out globally competent pioneering computing professionals, researchers, innovators and entrepreneurs and thereby succeed and contribute value to the knowledge-based economy and society."	1. To offer high-grade, value-based Post-graduate programme in the field of Computer Applications. 2. To develop the youth into professionals who can work in team, possess high analytical abilities, and help in solving complex problems of various domains through principles of computer science and applications.

SUBJECT: BLOCKCHAIN TECHNOLOGY

SUBJECT CODE: MC3T002

ASSIGNMENT ACTIVITY

1. Explain the fundamental architecture of Blockchain Technology and describe its major components.
2. Explain the concept of decentralization in blockchain. How does it differ from a centralized system?
3. Describe the structure of a blockchain block. Explain the significance of the block header and block body.
4. Explain cryptographic hashing in blockchain technology. Discuss the important properties of a secure hash function.
5. Explain the working of the Merkle Tree and discuss its importance in blockchain systems.
6. Describe the process of transaction verification and block creation in the Bitcoin blockchain.
7. Explain the Proof of Work consensus mechanism. Discuss its advantages, limitations, and security implications.
8. Explain the Proof of Stake consensus mechanism and compare its working with Proof of Work.
9. Discuss the different types of blockchain networks: public, private, consortium, and hybrid blockchain.
10. Explain the concept of digital signatures in blockchain. How do public and private keys contribute to transaction security?
11. Explain Bitcoin's blockchain architecture and describe the role of miners, nodes, wallets, and transactions.
12. What is a smart contract? Explain its characteristics, working mechanism, and applications.
13. Explain Ethereum architecture and discuss the role of Ether, accounts, transactions, and smart contracts.
14. Explain the concept of Byzantine Fault Tolerance (BFT). Why is it important for distributed blockchain networks?
15. Discuss the major security threats to blockchain systems, including the 51% attack, Sybil attack, and double-spending attack.
16. Explain the double-spending problem. How does blockchain technology prevent or minimize this problem?
17. Discuss the advantages and limitations of Blockchain Technology in real-world applications.

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18. Explain how blockchain can be used in supply-chain management. Discuss its benefits and challenges.
19. Discuss the applications of blockchain technology in areas such as healthcare, banking, voting, education, and IoT.
20. Explain the scalability, privacy, and interoperability challenges of blockchain technology. Discuss possible approaches for addressing these challenges.